Predictive Model Plan

1. Model Logic (Generated with GenAI)

The goal of the predictive model is to **forecast whether a customer is likely to become delinquent** based on financial and behavioral variables in the dataset. The model uses historical customer data such as credit utilization, income, missed payments, and debt-to-income ratio to identify patterns associated with delinquency risk.

Predictive Modeling Pipeline

Step 1: Data Input

- Load customer dataset containing financial, behavioral, and demographic variables.

Step 2: Feature Selection
Key features used for prediction include:

- Credit Score
- Credit Utilization
- Missed Payments
- Debt-to-Income Ratio
- Income
- Loan Balance
- Payment history (Month_1 – Month_6)

Step 3: Data Preprocessing

- Handle missing values using imputation (median or mean).
- Encode categorical variables (Employment Status, Credit Card Type, Location).
- Normalize numerical features such as income, credit utilization, and loan balance.
- Split the dataset into **training (80%) and testing (20%) sets**.

Step 4: Model Training

- Train a **Logistic Regression model** to classify customers into delinquent or non-delinquent categories.

Step 5: Prediction

- The model calculates a **probability score of delinquency** for each customer.
- If probability > threshold (e.g., 0.5), classify as **high risk**.

Simplified Pseudocode

```
Load dataset
Handle missing values
Encode categorical variables
Select key features

Split data into training and testing sets

Train Logistic Regression model on training data

For each customer in test data:
    Calculate probability of delinquency

If probability > threshold:
    Predict Delinquent
Else:
    Predict Non-Delinquent

Evaluate model performance using evaluation metrics
```

Model Summary

A **logistic regression model** is used to predict the probability that a customer will become delinquent. The model analyzes financial and behavioral variables to classify customers into delinquent or non-delinquent categories. This approach provides interpretable predictions that can support risk management decisions.

Top 5 Input Features

- **Missed Payments** – Indicates recent repayment behavior and financial reliability.
- **Credit Utilization** – High utilization suggests financial stress.
- **Debt-to-Income Ratio** – Measures how much income is committed to debt obligations.
- **Credit Score** – Reflects overall creditworthiness.
- **Income** – Helps assess a customer's capacity to repay debt.

2. Justification For Model Choice

Logistic regression was selected because it provides a strong balance between predictive performance and interpretability, which is particularly important in financial risk modeling. Unlike more complex models, logistic regression allows analysts to clearly understand how each variable influences the probability of delinquency. This transparency is essential for regulatory compliance and building trust in automated decision systems within financial institutions like Geldium. Additionally, logistic regression is computationally efficient, easy to deploy, and well suited for binary classification problems such as predicting whether a customer will become delinquent. While more complex models such as random forests or neural networks may provide slightly higher accuracy, logistic regression offers clearer explanations and easier monitoring, making it highly appropriate for operational risk management.

3. Evaluation Strategy

To evaluate the predictive model, several performance metrics will be used to assess accuracy, reliability, and fairness.

Key Evaluation Metrics

- **Accuracy** – Measures the overall percentage of correct predictions.
- **Precision** – Indicates how many predicted delinquent customers are actually delinquent.
- **Recall (Sensitivity)** – Measures how well the model identifies true delinquent customers.
- **F1 Score** – Provides a balance between precision and recall.
- **ROC-AUC** – Evaluates how well the model distinguishes between delinquent and non-delinquent customers.

Interpretation of Metrics

- A **high recall score** is important to ensure that most high-risk customers are correctly identified.
- **Precision** helps avoid falsely labeling low-risk customers as delinquent.
- **AUC scores closer to 1** indicate strong classification performance.

Bias and Fairness Checks

- Evaluate model predictions across demographic groups such as **age groups, employment status, or location**.
- Check whether the model systematically favors or disadvantages specific customer segments.
- Apply fairness monitoring techniques and adjust features if biased patterns are detected.

Ethical Considerations

- Ensure predictions are used to **support responsible lending decisions**, not unfairly deny services.
- Maintain transparency in how predictions are generated.
- Regularly monitor and update the model to prevent bias and maintain accuracy.

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